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Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree (R,S) Examination May 2024 (2019 Scheme)

Course Code: EET468

Course Name: INDUSTRIAL INSTRUMENTATION AND AUTOMATION

Max. Marks: 100 **Duration: 3 Hours**

PART A

Answer all questions, each carries 3 marks. Marks

1		Mention the factors influencing the choice of a transducer for Industrial applications.	(3)
2		Draw the step and ramp input responses for a first order sensor.	(3)
3		Explain Log amplifier, with relevant expressions.	(3)
4		Name any 6 signal conditioning circuits with its application.	(3)
5		Mention 3 advantages of Virtual instruments.	(3)
6		How does Profibus help in industrial data communication?	(3)
7		Implement two input AND and OR gate operations using PLC ladder logic.	(3)
8		Explain PLC ladder programming using a sample program.	(3)
9		Mention 3 features of DNP3 protocol.	(3)
10		What are the functions of RTU ?	(3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

11	a)	Draw and briefly explain the block diagram of process control loop with an example industrial process.	(7)
	b)	Explain the working of a Capacitive Differential Pressure Transducer. For what measurement is it commonly employed ?	(7)
		OR	
12	a)	Explain with block diagram analog and digital phase measurement techniques.	(7)
	b)	What is a hot-wire anemometer? Explain the principle and working of constant current type anemometer. Mention one advantage.	(7)

Module II

13	a)	How is a Differential Amplifier different from Instrumentation Amplifier? Derive the expression for Gain of an Instrumentation Amplifier.	(7)
	b)	Explain the operation of an electrical actuator. Mention two applications.	(7)

OR

14	a)	Explain the operation of Half wave and Full wave Precision rectifiers with circuit analysis.	(7)
	b)	Explain the operation of any two actuators with schematic diagram. Mention their industrial applications.	(7)

Module III

15	a)	Explain the architecture of a Virtual instrumentation system with block diagram.	(10)
	b)	Mention five functions of a Process control network in industry.	(4)

OR

16	a)	Compare Graphical programming with sequential programming. Where are Graphical programming employed ?	(8)
	b)	Explain any two different types of communication networks used for data collection and control in industry.	(6)

Module IV

17	a)	Explain two different types of Timers that may be used in a PLC.	(6)
	b)	Draw the block diagram architecture of a PLC and explain different sub components.	(8)

OR

18	a)	Explain operation of a Counter as used in a PLC. Mention two applications.	(6)
	b)	Draw a ladder logic program to switch on a pump for 100s. It is then to be switched off and a heater switched on for 50s. Then the heater is switched off, and another pump is used to empty the water.	(8)

Module V

19	a)	Explain the architecture of SCADA in process automation.	(7)
	b)	Compare SCADA and DCS for automation.	(7)

OR

20	a)	Explain the components of DCS architecture for industrial process monitoring and control.	(8)
	b)	Explain the role of HMI and MTU in SCADA communication.	(6)

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